

Physics-Informed Reconstruction of Unmeasurable Cardiac Fields from Real Data: Electrocardiographic Imaging and 4D-Flow Aortic Pressure, Verified Against Known Answers, with Null Results

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CAOS open-research programme · CAOS_RES_CardioPINN, physics-informed reconstruction of cardiac fields

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Not peer reviewed. Source code, data and computational records:

github.com/fsantibanezleal/CAOS_RES_CardioPINN

Abstract. Two clinically decisive cardiac quantities cannot be measured where they are needed: the heart-surface potential map (measured only with an invasive electrode cage) and the intra-vascular pressure field (measured only with a catheter). Each is tied by a governing equation to a quantity measurable non-invasively, the body-surface potential and the 4D-flow velocity, so each can be reconstructed. We report a real-data-only study of this reconstruction across two distinct physics domains under one discipline: fit only real measured signals; validate against a real gold standard where one exists and against an exact analytic known-answer test where none can; and report the null results. The common methodological spine is to separate the well-posed part that data constrains from the ill-posed part that the physics forces out, and to verify every physics engine on a closed-form problem before it is applied to real data. *Case 1 (electrocardiographic imaging, quasi-static volume conduction).* A forward operator on the real geometry with a graph-Laplacian Tikhonov prior and a deep-ensemble node uncertainty recovers heart-surface potentials from real EDGAR experiments by an identical pipeline with no per-heart retuning, at relative error 0.54–0.65 and spatial correlation 0.72–0.85 against the measured cage, with about 90% of nodes inside two standard deviations, and the paced beats reconstructing better than sinus as physically expected. A full boundary-element forward operator, verified to correlation above 0.999 on an analytic concentric-sphere problem, does *not* beat the calibrated single-layer on the real coarse electrode geometry (dog: 0.542 vs 0.629 relative error), a null result. *Case 2 (4D-flow aortic pressure, incompressible Navier–Stokes).* A divergence-free velocity network denoises a real thoracic-aorta 4D-flow scan (raw divergence reduced $2.3\times$) and the relative pressure is forced out by the pressure-Poisson equation with the source and Neumann flux computed from the network’s analytic derivatives rather than by finite differences at the lumen edge, which takes the recovered pressure from a non-physiological thousands of mmHg to a physiological 0.79 mmHg range, the same order as the clinical simplified-Bernoulli

estimate (2.50 mmHg). The solve is verified to correlation 1.00 on an analytic converging duct, and its unsteady term to dw/dt correlation 0.995 on a time-varying Poiseuille flow; the momentum-residual PINN, which leaves pressure gauge-free, serves as a failed baseline. The uncertainty treatment is case-appropriate: informative and shown per node where the inverse is ill-posed, and summarized as a scalar where the denoiser makes it near-zero. The contribution is a demonstration that, with a real-data-only discipline and known-answer verification, physics-informed reconstruction recovers two clinically meaningful unmeasurable cardiac fields from the non-invasive measurements tied to them.

Index Terms. physics-informed neural networks, electrocardiographic imaging, inverse problems, 4D-flow MRI, pressure-Poisson equation, Navier–Stokes, volume conduction, boundary element method, deep ensembles, uncertainty quantification, analytic verification, cardiac reconstruction.

I. INTRODUCTION

PHYSICS-INFORMED neural networks (PINNs) embed a governing differential equation in the training loss, so a network can fit sparse or noisy measurements while respecting the physics that relates them to unobserved fields [1]. Their most compelling use is not re-solving a forward problem a classical solver already handles, but recovering a field that *cannot* be measured from one that can [2]. Two cardiac problems have exactly this shape. In electrocardiographic imaging (ECGi), the clinician needs the heart-surface potential map to localize an arrhythmia, but a patient offers only the body-surface recording; the true heart-surface map, from an electrode cage, is a gold standard one never has in the clinic [5]. In vascular flow, the clinician needs the pressure drop across a stenosis, but 4D-flow MRI measures only velocity, and pressure is obtained invasively by catheter [4].

This paper is deliberately real-data-only. A recurring failure mode in physics-informed reconstruction is to validate on synthetic data, where a network is scored against a field a solver produced under the same assumptions the network uses; such a result demonstrates code correctness, not a clinical capability. We therefore fit only real measured signals and

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Code, derived reconstruction artifacts, and the interactive viewer are released under MIT at https://github.com/fsantibanezleal/CAOS_RES_CardioPINN; the viewer is available at <https://cardiopinn.fasl-work.com>. Validated methodological result on real experimental data, not a clinically deployed system; raw datasets are used under their data-use agreements and are not redistributed.

validate only against a real measured gold standard or, where none can exist, against an exact analytic test that must pass before any real datum is trusted. We report two cases in two different physics domains and, crucially, the negative results each produced: a boundary-element forward operator that does not beat a simpler one on real geometry, and a momentum-residual PINN that cannot recover pressure at aortic Reynolds number. The contributions are (i) a shared methodological spine, with known-answer verification of every physics engine, for recovering an unmeasurable clinical field from a measurable one; (ii) a real-data ECGi reconstruction validated per beat against the measured cage, with the boundary-element null; and (iii) a real-data 4D-flow pressure reconstruction validated on two analytic tests and bracketed by the clinical Bernoulli estimate, with the momentum-residual failure documented.

II. THE COMMON METHODOLOGICAL SPINE

Both cases share one spine (Fig. 1). Each separates a well-posed part that data constrains (the body-surface potentials; the velocity field) from an ill-posed part that the physics forces out (the heart-surface potentials via a regularized inverse; the pressure via an elliptic equation). And each verifies its physics engine on a known-answer analytic problem before any real data are used: an engine that produces a plausible-looking field on real data proves nothing, because it can be wrong in a way no real measurement exposes. The test must pass before the real reconstruction is trusted. This requirement is what excluded the momentum-residual PINN of Section IV, which failed its analytic Poiseuille test.

III. CASE 1: ELECTROCARDIOGRAPHIC IMAGING

A. Forward operator and regularized inversion

In the quasi-static approximation the torso is a volume conductor and the potential ϕ obeys Laplace's equation $\nabla \cdot (\sigma \nabla \phi) = 0$ between the heart surface and the body surface [8]. Discretized on the real geometry, the body-surface potentials b are a linear map of the heart-surface potentials x , $b = Ax + \varepsilon$, where A is the transfer matrix and ε measurement noise. The singular values of A decay rapidly, so recovering x is severely ill-posed. We build A two ways: a single-layer point-source Green's-function approximation, and a full boundary-element operator (BEM) whose double-layer entries use the exact solid angle of a plane triangle [9]. The reconstruction is a Tikhonov problem with a graph-Laplacian prior L on the heart surface,

$$\hat{x} = \arg \min_x \|Ax - b\|_2^2 + \lambda \|Lx\|_2^2, \quad (1)$$

with λ chosen on the L-curve [10]. Because a single point estimate carries no error bar, we draw a deep ensemble over realistic measurement-noise perturbations of b and resolve [11], and recalibrate its two-standard-deviation band to cover the true per-node error at the nominal rate.

B. The concentric-sphere analytic test

The BEM is verified on two concentric spheres, an inner heart sphere of radius $a = 0.7$ and an outer insulating body

sphere $b = 1.0$. For a heart-surface potential equal to the degree-one harmonic $\phi_H = \cos \theta$, the shell solution $\phi = (Ar + B/r^2) \cos \theta$ with the insulating condition $\partial \phi / \partial r = 0$ at $r = b$ gives, in closed form,

$$B = \frac{1}{2}Ab^3, \quad Aa + \frac{B}{a^2} = 1, \quad \frac{\phi_B}{\cos \theta} = Ab + \frac{B}{b^2}, \quad (2)$$

so the exact heart-to-body transfer ratio is known. The BEM reproduces it at correlation above 0.999, magnitude within 8%, and first-order convergence (the error halves per mesh refinement, the signature of a correct assembly rather than an accidental match). Only then is it applied to the one real case whose two surfaces are closed 2-manifolds, the dog.

C. Real validation on EDGAR and the boundary-element null

The pipeline is applied, with no per-heart retuning, to independent real EDGAR experiments [6]: a human torso tank (192 body $\rightarrow$ 256-node cage; sinus and two paced beats) and an in-situ dog (140 body $\rightarrow$ 1321-node epicardium; sinus). Recovered potentials are compared to the measured cage with the standard relative error (RE) and spatial correlation (CC) [7] (Fig. 2). The numbers are literature-consistent for torso-tank ECGi, the node uncertainty covers about 90% of nodes at two standard deviations, and the paced beats reconstruct better than sinus (CC rising from 0.72 at sinus to 0.80 and 0.85 paced), which is physically expected: a focal paced activation is easier to localize than a diffuse sinus wavefront.

A more physically complete forward operator is not automatically a better one. The BEM passes its analytic test, yet on the real electrode geometry it does not beat the calibrated single-layer: it requires closed 2-manifold surfaces (so it applies only to the dog case), and where it applies the coarse 140-node torso makes the reconstruction regularization-dominated, single-layer relative error 0.542 versus BEM 0.629. Forward-operator fidelity is not the bottleneck at this electrode density; the single-layer stays the default, and the BEM is expected to matter only as electrode density and mesh closure improve.

IV. CASE 2: 4D-FLOW AORTIC PRESSURE

A. The pressure-Poisson route

For an incompressible Newtonian fluid the momentum equation is $\rho(\partial_t v + (v \cdot \nabla)v) = -\nabla p + \mu \nabla^2 v$ with $\nabla \cdot v = 0$. Taking the divergence and using incompressibility yields a Poisson equation for pressure,

$$\nabla^2 p = S(v) = -\rho \nabla \cdot ((v \cdot \nabla)v) + \rho \nabla \cdot (\partial_t v), \quad (3)$$

whose source is built from spatial derivatives of the measured velocity. Because that source is a product of derivatives, measurement noise (which violates incompressibility) is amplified, so a physics-informed velocity step is required first: a network fits the measured velocity while enforcing $\nabla \cdot v = 0$,

$$\mathcal{L} = \|v_\theta - v_{\text{meas}}\|_2^2 + \beta \|\nabla \cdot v_\theta\|_2^2, \quad (4)$$

producing a smooth divergence-free field (raw divergence reduced 2.3 $\times$, from 25.4 to 11.2 s⁻¹) whose analytic derivatives are clean. The source and Neumann wall flux in (3)

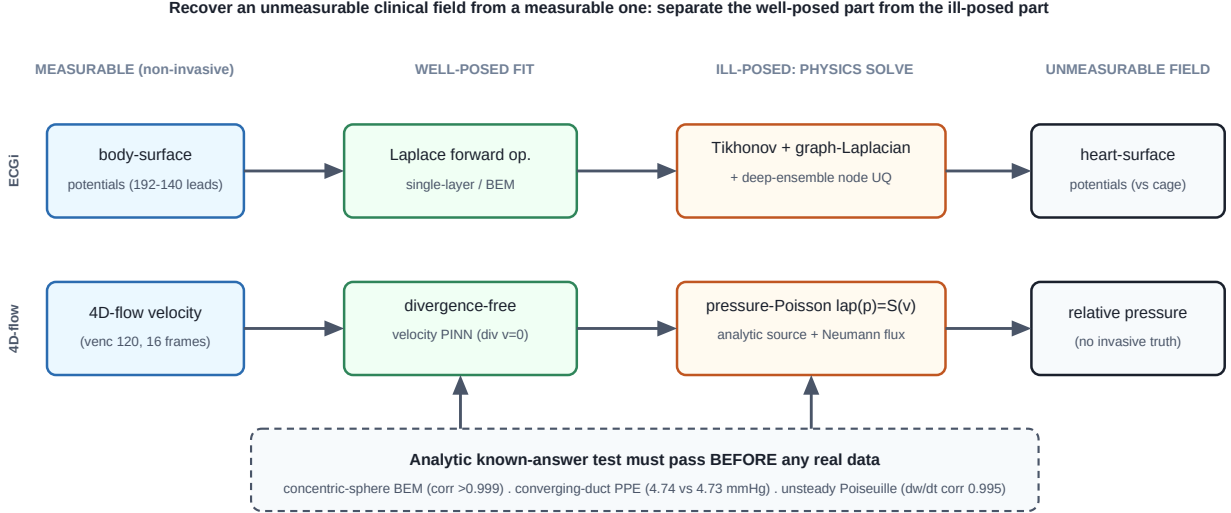


Fig. 1. The shared method. In each domain a non-invasive measurable field constrains a well-posed fit; the unmeasurable clinical field is then forced out by the ill-posed physics step (a regularized inverse for ECGi, an elliptic pressure solve for 4D-flow); and every physics engine must first reproduce an exact answer on a closed-form problem before it is applied to real data.

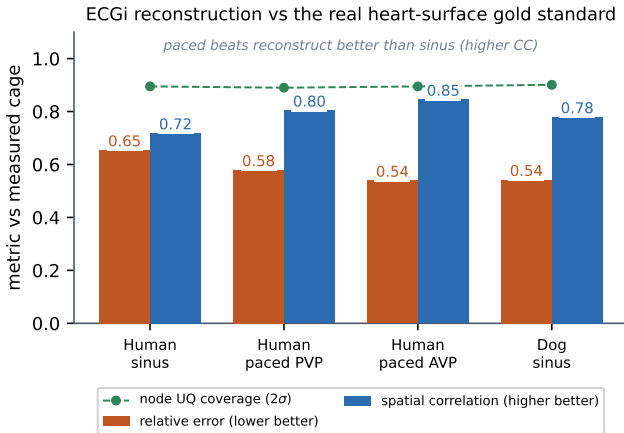


Fig. 2. Real ECGi validation against the measured cage potentials (identical pipeline, no per-heart retuning). Relative error (orange, lower better) and spatial correlation (blue, higher better) per beat, with the node uncertainty coverage at two standard deviations (green). Paced beats reconstruct better than sinus. Data: catalogue.json.

are computed from those analytic derivatives, not by finite differences at the lumen edge where they manufacture the worst artifacts [3]; the unsteady term $\partial_t v$ is differentiated exactly by a space-time network trained over the whole cardiac cycle. A single network trained on the raw momentum residual (the hidden-fluid-mechanics formulation [2]) does not recover pressure at aortic Reynolds number: pressure is gauge-free and only weakly coupled to the loss, so it stays near its initialization (under 10% of the true gradient on analytic Poiseuille). That approach is retained as a failed baseline; the adopted method separates the well-posed velocity from the ill-posed pressure.

TABLE I

THE ANALYTIC KNOWN-ANSWER TESTS. EACH ENGINE REPRODUCES AN EXACT CLOSED-FORM ANSWER BEFORE IT IS APPLIED TO REAL DATA.

Test	Analytic problem	Recovered vs true
BEM (ECGi)	concentric spheres	corr > 0.999; err halves/refine
PPE steady	converging duct	corr 1.00; 4.74 vs 4.73 mmHg
PPE unsteady	time-varying Poiseuille	dw/dt corr 0.995

B. Two analytic tests

The steady solver is verified on an analytic converging duct with a mass-conserving field $w = U_0(1 + az)$, $u = -\frac{a}{2}U_0x$, $v = -\frac{a}{2}U_0y$, whose axial Euler pressure is known,

$$p(z) = -\rho U_0^2 \left(az + \frac{a^2 z^2}{2} \right) + c, \quad (5)$$

recovered at correlation 1.00 and a pressure drop of 4.74 mmHg against the analytic 4.73. The unsteady term is verified on a time-varying Poiseuille flow $w(r, t) = U_0(1 + A \sin \omega t)(1 - (r/R)^2)$, whose exact axial acceleration on the axis ($r = 0$), $\partial_z p|_{\text{unsteady}} = -\rho \partial_t w|_{r=0} = -\rho U_0 A \omega \cos \omega t$, is recovered there at correlation 0.995. These tests are what license the analytic unsteady term on the real scan instead of a noisy three-frame finite difference; the consequence is direct (the finite-difference term inflates the recovered range to 14.87 mmHg, the analytic term gives 0.79). The three tests are summarized in Table I.

C. Real validation with no invasive gold standard

There is no non-invasive pressure gold standard; the validation is threefold. The analytic test is exact (above). On the real thoracic-aorta scan (Philips, venc 120 cm/s, 16 frames, 47,902 lumen voxels, 27,863 phase-wrap samples corrected), the recovered relative pressure has a 2.5–97.5 percentile range

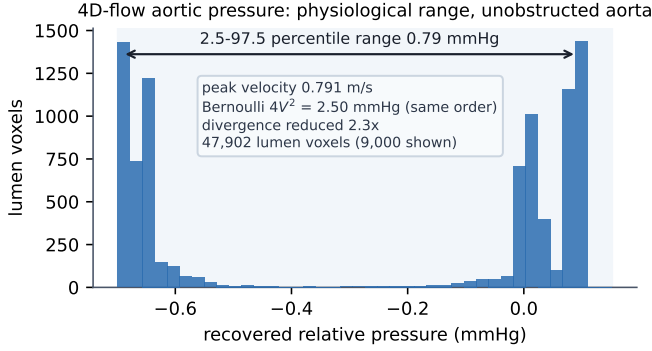


Fig. 3. Recovered relative pressure over the aortic lumen at peak systole. The distribution over the 9,000 lumen voxels sampled for the released trace has a 2.5–97.5 percentile range of 0.79 mmHg (arrow) and a full span of 0.85 mmHg (shaded) for this unobstructed aorta, the same order as the clinical Bernoulli estimate (2.50 mmHg). Data: `trace.json`.

of 0.79 mmHg over the 9,000 lumen voxels sampled for the released trace (full span 0.85 mmHg; Fig. 3), small and physiological for an unobstructed aorta. The simplified-Bernoulli estimate $4V_{\max}^2 = 2.50$ mmHg, computed from the denoised peak velocity 0.791 m/s (the maximum speed of the denoised velocity field over the lumen), is of the same order. The measured speed, lightly smoothed, reaches about 1.44 m/s in the sampled voxels, which would give about 8.3 mmHg; the agreement therefore holds for the denoised peak velocity and not for the raw measured peak. The field additionally shows where the pressure varies, which a single Bernoulli number cannot.

A deep ensemble that perturbs the measured velocity with realistic phase-contrast noise (5% of the v_{enc}) and re-runs the whole pipeline moves the recovered pressure by under 0.01 mmHg: the divergence-free denoiser makes the pressure essentially insensitive to velocity measurement noise. The same robustness means that such an ensemble gives a near-zero, uninformative uncertainty, so we report the robustness as a scalar and deliberately do not show a per-voxel uncertainty map, which would be a misleadingly uniform near-zero field. This is the opposite of Case 1, where the ill-posed inverse makes the ensemble uncertainty large and informative, so it is shown per node.

V. DISCUSSION

The two cases share one methodological spine and differ exactly where the physics differs. Each separates a well-posed part that data constrains from an ill-posed part that the physics forces out; each validates against something real, a measured gold standard where one exists (the ECGi cage) and an exact analytic test where one cannot (the converging-duct pressure drop); and each has a negative result, the BEM that does not beat the single-layer on real coarse geometry and the momentum-residual PINN that leaves pressure gauge-free. The uncertainty treatment is case-appropriate rather than uniform: informative and shown per node where the inverse is ill-posed, and summarized as a scalar where the denoiser makes it near-zero. The null results and the uncertainty matched to the

problem are what separate a validated methodological result from a demonstration.

VI. LIMITATIONS AND CONCLUSION

These are validated methodological results on real experimental data, not clinically deployed tools. ECGi is shown on a torso tank and one in-situ animal, not a patient cohort; the 4D-flow pressure has no invasive truth to check its absolute magnitude against, so only its analytic verification, physiological range, divergence-free denoising, and Bernoulli bracket are claimed. Raw datasets are used under data-use agreements and are not redistributed; only the derived reconstruction artifacts are released, and every number here is reproducible from them and from the automated tests in the repository. The contribution is a demonstration that, with a real-data-only discipline and known-answer verification, physics-informed reconstruction can recover two clinically meaningful unmeasurable cardiac fields, ECGi heart-surface potentials and 4D-flow aortic pressure, from the non-invasive measurements tied to them by a governing equation.

DATA AND CODE AVAILABILITY

Code, the derived reconstruction artifacts (the ECGi catalogue and the 4D-flow pressure trace), and the interactive viewer are at

https://github.com/fsantibanezleal/CAOS_RES_CardioPINN under the MIT license; a static viewer is available at <https://cardiopinn.fasl-work.com>. ECGi data are from the EDGAR repository (Consortium for ECG Imaging); the 4D-flow scan is used under its data-use agreement. Raw datasets are not redistributed. The figures are regenerated deterministically from these artifacts by `manuscripts/physics-informed-cardiac-fields/figures/make_figs.py`.

APPENDIX A PER-BEAT ECGi REGULARIZERS

The reported ECGi metrics use the single-layer Tikhonov reconstruction. The catalogue also records a graph-regularized and a deep-ensemble variant per beat; on the human tank the single-layer relative error / correlation are 0.654/0.718 (sinus), 0.579/0.804 (paced PVP), and 0.539/0.845 (paced AVP), with node-uncertainty two-standard-deviation coverage 0.895/0.890/0.895; the dog sinus case is 0.542/0.779 (single-layer) versus 0.629/0.775 (BEM). All per-beat, per-regularizer values are available in `catalogue.json`.

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